Mach-Zehnder Interferometer Design, Fabrication and Data Analysis

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***Abstract*—This report outlines the design and characterisation of silicon photonics Mach-Zehnder interferometer (MZI) on a 220 nm Silicon on Insulator (SOI) platform. Theoretical modelling, simulations, fabrication and data analysis are used to understand and analyse the transmission response of multiple imbalanced MZI.**

***Index Terms*—Group Index, Group Delay, Free spectral range, Mach-Zehnder Interferometer, Silicon photonics, Transmission Response, Waveguide, Y branch Splitter/Coupler**

I. INTRODUCTION

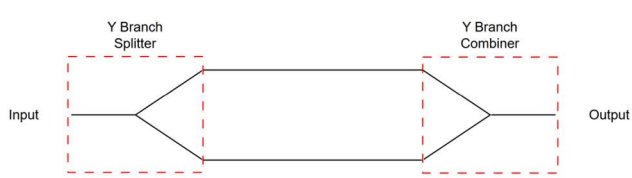
SILICON Photonics enables integration of optical components onto silicon chips using standard semiconductor fabrication process. The Mach-Zehnder interferometer (MZI) is a key building block in silicon photonics and is widely used in devices such as optical switches and modulators. An MZI in this case is works by splitting optical light into two paths, introducing a phase difference between the branches and recombine them to produce constructive or destructive interference.

In this report we focus on the design and characterization of an MZI. The behaviour of these components is simulated using the Lumerical simulation software and validated using fabrication and measurement analysis.

II. THEORY

An MZI in silicon photonics can be constructed using three main optical components:

1. Waveguides
2. Grating Couplers
3. Y branch splitters/combiners



Let's explore each individual component:

Waveguide

Wave guides are fundamental blocks in silicon photonics. They work by confining and guiding the light through the silicon layer. We will design waveguide with silicon core, which has high refractive index, and is surrounded by silicon dioxide cladding with low refractive index.

Light propagation in a silicon waveguide can be described using electromagnetic wave theory

$$E = E_0 \cdot e^{j(\omega t - \beta z)}$$

Where:

$Angular\ frequency\ (\omega) = \frac{2\pi c}{\lambda}$,

$Propogation\ constant\ (\beta) = \frac{2\pi n_{\text{eff}}}{\lambda}$ and

n_{eff} is the effective refractive index.

Y Branch Splitters and Combiners:

Y branches can be used to either split the light or recombine them. Let's say, light with intensity I_i and electric field E_i enters a Y-branch splitter, it gets equally divided between the two output branches and this can be expressed as

$$I_1 = I_2 = \frac{I_i}{2}$$

Since the intensity is proportional to the square of the electric field magnitude

$$I \propto |E|^2$$

$$|E_1|^2 = |E_2|^2 = \frac{|E_i|^2}{2}$$

Therefore, the electric field amplitudes are:

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}}$$

When light from two input waveguides with electric fields E_1 and E_2 enters a Y-branch combiner the output electric field becomes a vector superposition of the inputs, scaled by $1/\sqrt{2}$:

$$E_{out} = \frac{1}{\sqrt{2}}(E_1 + E_2)$$

The output intensity depends on both the magnitude and phase relationship between E_1 and E_2:

$$I_{out} = |E_{out}|^2 = \frac{1}{2}|E_1 + E_2|^2$$
$$= \frac{1}{2}(|E_1|^2 + |E_2|^2 + 2\text{Re}(E_1 E_2^*))$$

If both inputs have equal intensity ($|E_1| = |E_2| = E_0$):

$$I_{out} = \frac{1}{2}(E_0^2 + E_0^2 + 2E_0^2 \cos(\phi)) = E_0^2(1 + \cos(\phi))$$

where ϕ is the phase difference between E_1 and E_2.

This yields two extreme cases: Constructive Interference where we see maximum transmission and Destructive Interference, where we see complete extinction.

Grating couplers:
Grating couplers enable light coupling between external optical fibres to on chip waveguides.

Mach-Zehnder Interferometer Transfer Function
For an imbalanced MZI with identical waveguides ($\beta_1 = \beta_2 = \beta$) but different arm lengths, light propagating through each arm accumulates different phases. After propagating through arms of length L_1 and L_2, the fields at the end of each arm are:

$$E_{o1} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_1}, E_{o2} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_2}$$

The combiner output becomes:

$$E_{out} = \frac{1}{\sqrt{2}}(E_{o1} + E_{o2}) = \frac{E_i}{2}(e^{-i\beta L_1} + e^{-i\beta L_2})$$

The output intensity in the lossless case is:

$$I_{out} = |E_{out}|^2 = \frac{I_i}{4}|e^{-i\beta L_1} + e^{-i\beta L_2}|^2$$

After trigonometric simplification, the normalized transmission spectrum becomes:

$$T(\lambda) = \frac{I_{out}}{I_{in}} = \frac{1}{2}[1 + \cos(\beta(\lambda).\Delta L)]$$

$$T(\lambda) = \frac{1}{2}\left[1 + \cos\left(\frac{2\pi n_{eff}(\lambda)}{\lambda} \cdot \Delta L\right)\right]$$

where $\Delta L = L_1 - L_2$ is the path length imbalance.

Free Spectral Range and Group Index Extraction
The free spectral range (FSR) is the wavelength spacing between adjacent transmission peaks, defined as

$$\text{FSR} = \lambda_{m+1} - \lambda_m.$$

Given the imbalanced interferometer transfer function

$$\frac{I_o}{I_i} = \frac{1}{2}(1 + \cos(\beta\Delta L))$$

we define the phase term as $\beta\Delta L$, where the propagation constant is

$$\beta = \frac{2\pi n}{\lambda}$$

The phase difference between adjacent peaks is 2π. At adjacent peaks, the propagation constants are β_m and β_{m+1}, giving

$$(\beta_m - \beta_{m+1})\Delta L = 2\pi$$

Approximating β as varying linearly over the small wavelength step FSR, we use

$$\beta_m - \beta_{m+1} \approx \frac{d\beta}{d\lambda}\,\text{FSR}$$

Combining equations above yields

$$\text{FSR} = \frac{2\pi}{\Delta L\,\frac{d\beta}{d\lambda}}$$

Using $\beta = \frac{2\pi}{\lambda}$, the derivative is

$$\frac{d\beta}{d\lambda} = \frac{2\pi}{\lambda^2}\left(\lambda\frac{dn}{d\lambda} - n\right)$$

Substituting equation into equation gives

$$\text{FSR} = \frac{\lambda^2}{\Delta L\left(n - \lambda\frac{dn}{d\lambda}\right)} = \frac{\lambda^2}{\Delta L\, n_g}$$

where the group index is defined as

$$n_g = n - \lambda\frac{dn}{d\lambda}$$

Rearranging equation, the group index can be extracted from a measured FSR and known imbalance length ΔL via

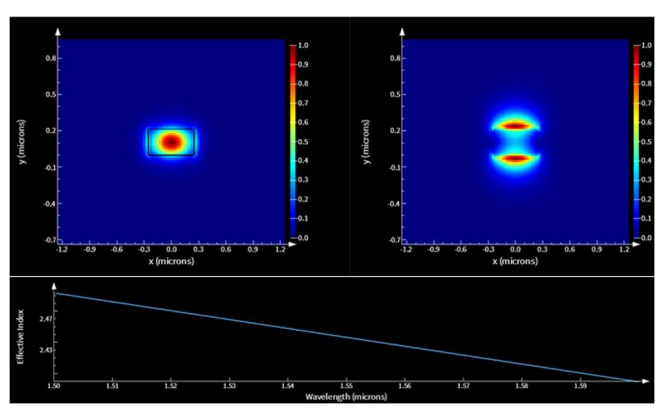
$$n_g = \frac{\lambda^2}{\Delta L \cdot \text{FSR}}$$

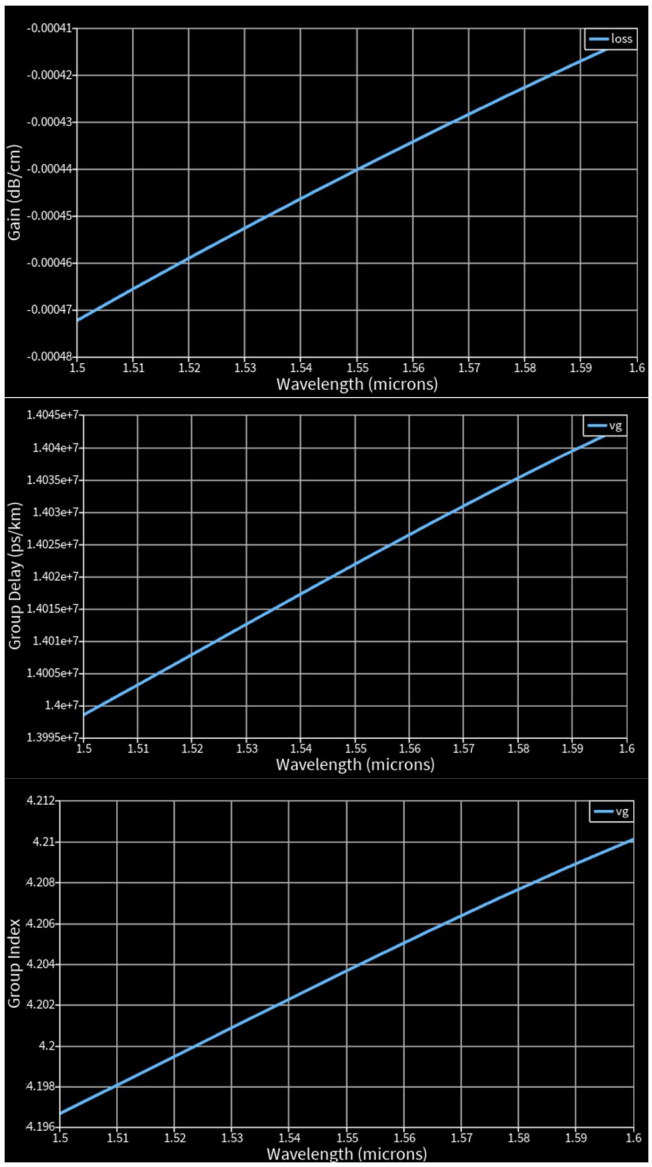
III. Modelling and Simulations

Design parameters:
The following design methodology was implemented. A total of nine Mach–Zehnder Interferometers (MZIs) were designed using the layout software KLayout. Of these, five MZI's have an identical path length difference of 100 um, two with 150 um and one with 950 um. We have also designed one with 20 um for baseline correction. This approach enables verification of measurement repeatability and assessment of design variability. The designs were constrained to an allocated layout area of 605 × 401 µm².

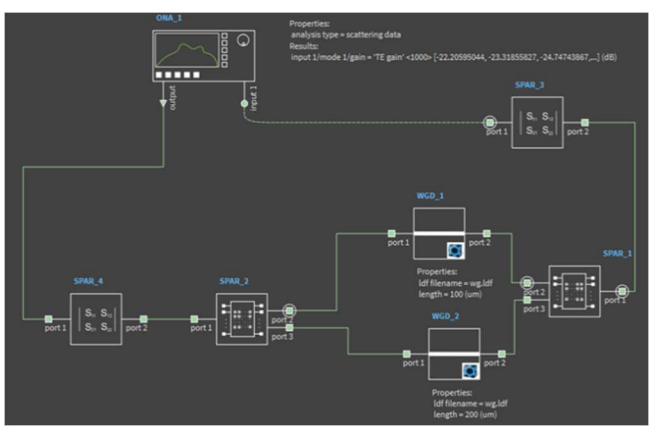
The MZIs were designed for operation at a centre wavelength of 1550 nm, with a wavelength sweep spanning 1500–1600 nm. A strip waveguide configuration was adopted, and only quasi-TE mode MZIs were considered. All waveguides were designed with a uniform cross section of 220 × 500 nm.
The effective refractive index and group index of the waveguides were calculated using the eigenmode solver in Lumerical MODE Solutions. The material refractive indices used in the simulations were n(Si) = 3.47 and n(SiO_2) = 1.44.

Let's simulate the waveguide in Lumerical mode simulation software.
The silicon waveguide was simulated using Lumerical MODE Solutions to evaluate its optical properties. The simulation results include the electromagnetic energy density profiles for both quasi-TE and quasi-TM modes, confirming single-mode operation for the target geometry. The effective refractive index was calculated as a function of wavelength across the operating band. In addition, key dispersive parameters including gain, group delay, and group index were extracted to support Mach-Zehnder Interferometer modelling and free spectral range analysis.

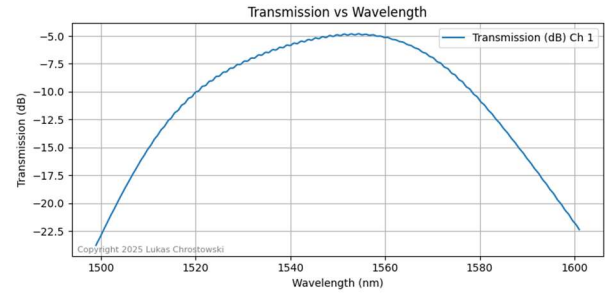




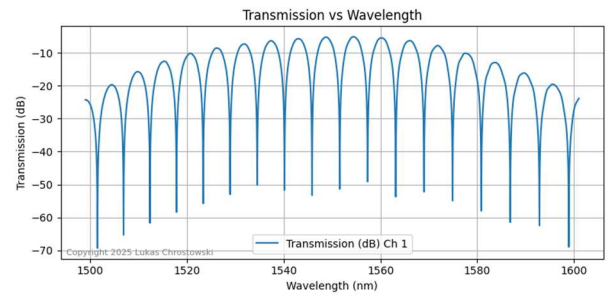
Then we simulated, imbalanced MZI's in Lumerical Interconnect and transmission plots are displayed below:



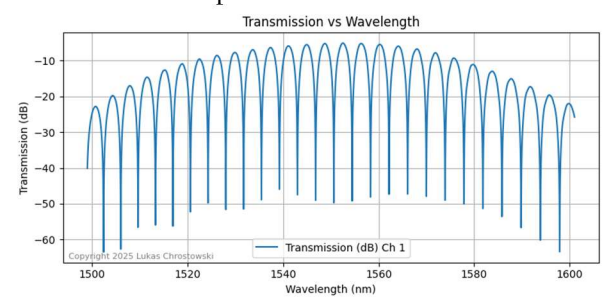
A branch length difference ranging from 20 µm to 950 µm was considered. A wavelength sweep was performed and the transmission vs wavelength data was plotted.



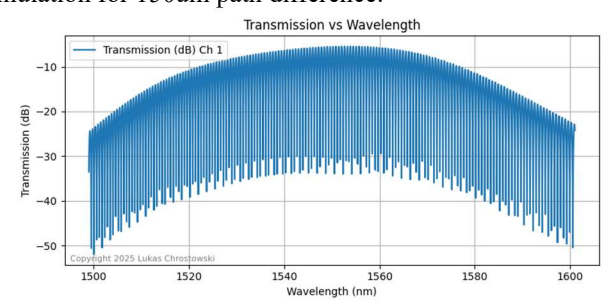
Simulation for 20 um path difference. This was designed to use for baseline calibration.



Simulation for 100um path difference.



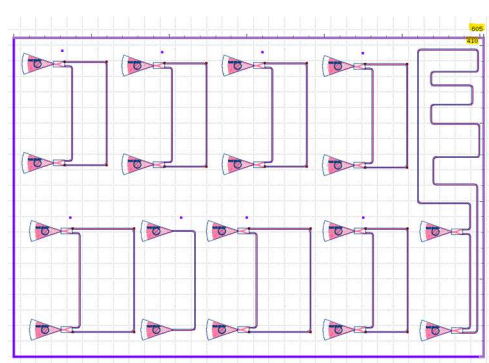
Simulation for 150um path difference.



Simulation for 950um path difference.

Design in Klayout

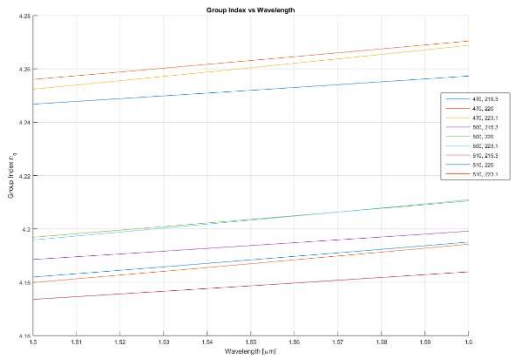
The designs were then created on Klayout.



Corner Analysis

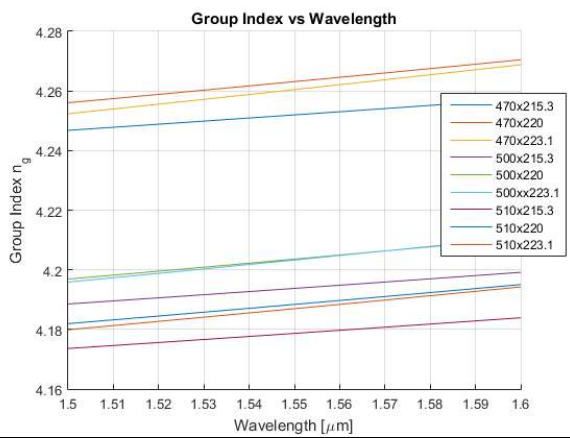
Corner analysis was then performed to determine the impact of the variations. Nine corners were simulated and the group index and FSR was calculated. We can find the results below:

Group index corner analysis for 100 um path length difference:



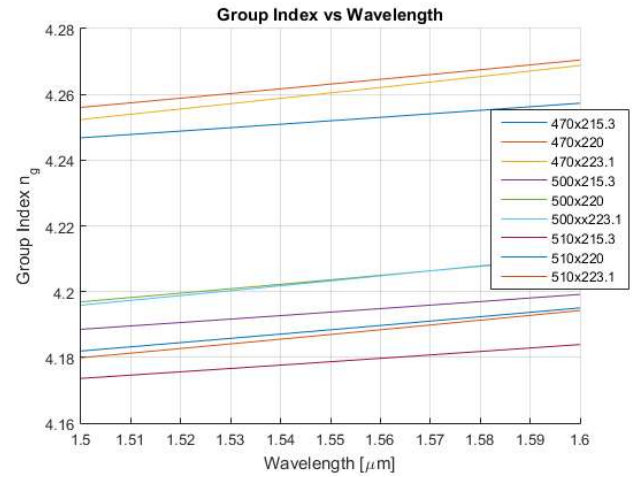
Waveguide size	ng	FSR (nm)
470 × 215.3	4.252	5.6503
470 × 220.0	4.263	5.6355
470 × 223.1	4.26	5.6391
500 × 215.3	4.194	5.7288
500 × 220.0	4.204	5.7154
500 × 223.1	4.203	5.7158
510 × 215.3	4.179	5.7494
510 × 220.0	4.188	5.7361
510 × 223.1	4.187	5.7381

Group index corner analysis for 150 um path length difference:



Waveguide size	ng	FSR (nm)
470 × 215.3	4.252	3.7669
470 × 220.0	4.263	3.757
470 × 223.1	4.26	3.7594
500 × 215.3	4.194	3.8192
500 × 220.0	4.204	3.8102
500 × 223.1	4.203	3.8105
510 × 215.3	4.179	3.833
510 × 220.0	4.188	3.8241
510 × 223.1	4.187	3.8254

Group index corner analysis for 950 um path length difference:



Waveguide size	ng	FSR (nm)
470 × 215.3	4.252	0.5948
470 × 220.0	4.263	0.5932
470 × 223.1	4.26	0.5936
500 × 215.3	4.194	0.603
500 × 220.0	4.204	0.6016
500 × 223.1	4.203	0.6017
510 × 215.3	4.179	0.6052
510 × 220.0	4.188	0.6038
510 × 223.1	4.187	0.604

IV. Fabrication

Summary of the fabrication description:

Fabricated is performed at one or more of these: Applied Nanotools and Washington Nanofabrication Facility. The following are the process descriptions.

Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam

drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4” handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

Washington Nanofabrication Facility (WNF) silicon photonics process

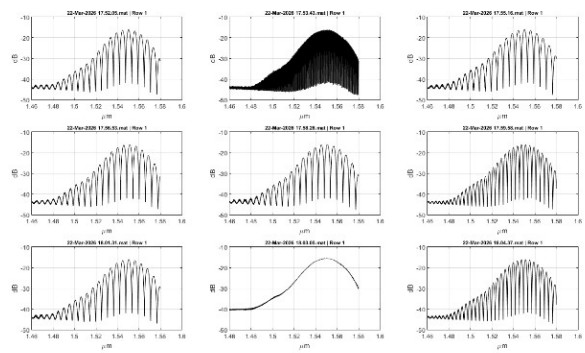
The devices were fabricated using 100 keV Electron Beam Lithography [1]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 µm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 °C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 µm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 µC/cm2 was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20 °C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil. Cladding oxide was deposited using plasma enhanced chemical vapor deposition (PECVD) in an Oxford Plasmalab System 100 with a silane (SiH4) flow of 13.0 sccm, nitrous oxide (N2O) flow of 1000.0 sccm, high-purity nitrogen (N2) flow of 500.0 sccm, pressure at 1400mT, high-frequency RF power of 120W, and a platen temperature of 350C. During deposition, chips rest directly on a silicon carrier wafer and are buffered by silicon pieces on all sides to aid uniformity.

Measurement description:

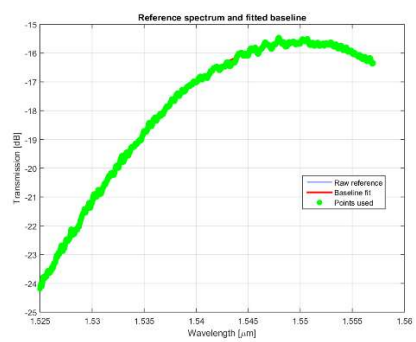
To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90º rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

V. Experimental data

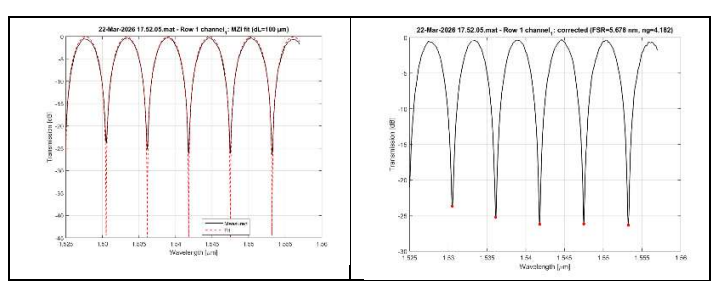
The plot below shows the experimental data measured using the custom-built automated measurement setup. The measured MZI transmission spectra were processed in MATLAB using a baseline-correction and fitting procedure. First, a reference dataset was used to estimate the spectral envelope by fitting a fourth-order polynomial to the top 10 dB of the spectrum within the wavelength range of 1525 - 1557 nm. This baseline was then subtracted from all measured spectra and re-normalized so that the peak transmission levels were aligned. This step removes the slowly varying coupling response and highlights the MZI interference fringes. After baseline correction, each spectrum was fitted using a nonlinear least-squares method to an analytical MZI transfer-function model, allowing key waveguide and interferometer parameters to be extracted.



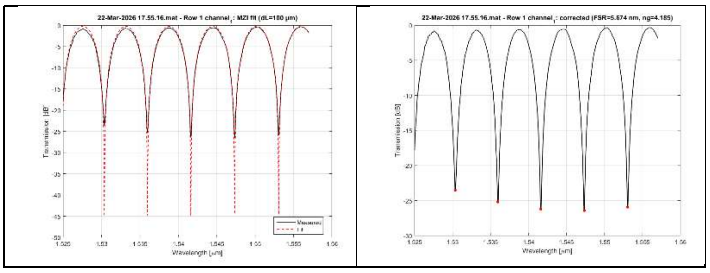
Baseline data



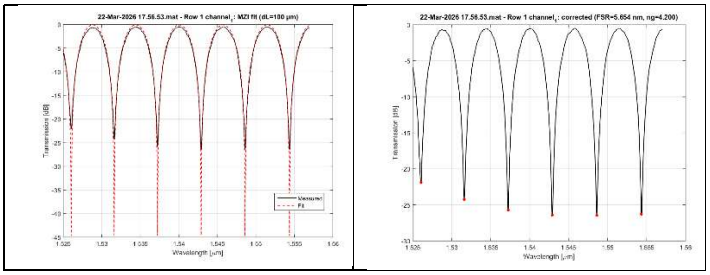
Path length difference of 100um – experimental data MZI 1



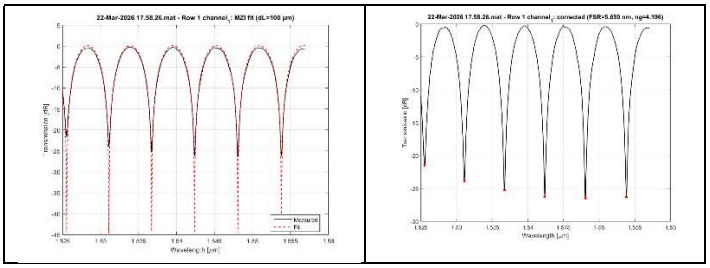
MZI 2



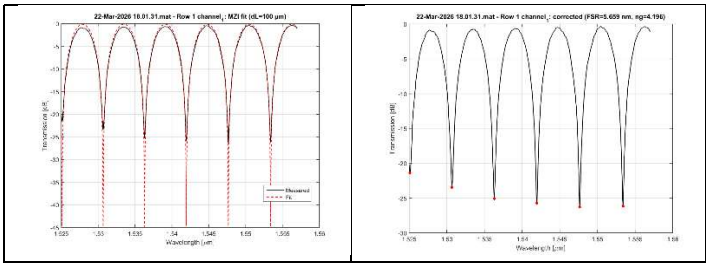
MZI3



MZI4

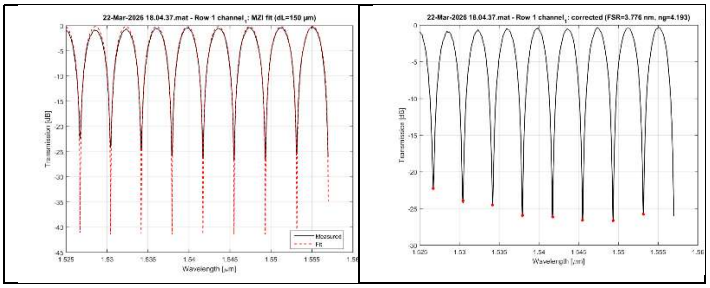


MZI 5

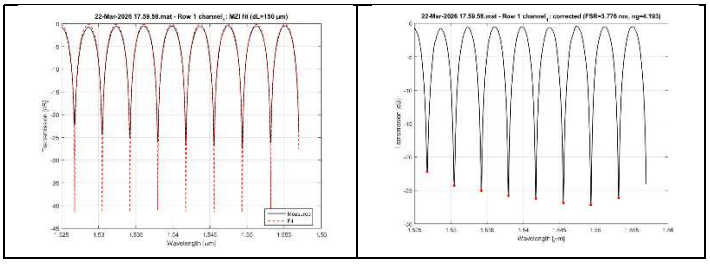


Path length difference of 150um – experimental data

MZI 6

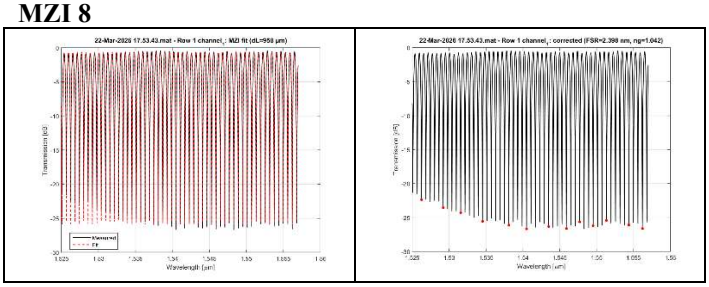


MZI 7



Path length difference of 950um - experimental data

MZI 8



Based on the corner analysis, we calculate fsr and ng min and max as follows:

For 100 um path difference -

Parameter	Minimum value	Maximum value
ng	4.179	4.263
FSR	5.6355	5.7494

For 150 um path difference -

Parameter	Minimum value	Maximum value
ng	4.179	4.263
FSR	3.757	3.833

For 950 um path difference -

Parameter	Minimum value	Maximum value
ng	4.179	4.263
FSR	0.5932	0.6052

Experimental data:

file	deltaL_um	FSR	ng
22-Mar-2026 17.52.05.mat	100	5.678	4.184447962
22-Mar-2026 17.55.16.mat	100	5.674	4.189544166
22-Mar-2026 17.56.53.mat	100	5.6544	4.187165182
22-Mar-2026 17.58.26.mat	100	5.6592	4.18874691
22-Mar-2026 18.01.31.mat	100	5.6592	4.187600482
22-Mar-2026 17.59.58.mat	150	3.776	4.185366829
22-Mar-2026 18.04.37.mat	150	3.776	4.183851185
22-Mar-2026 17.53.43.mat	950	0.5988	4.173289

Based on the results all designs are within the simulation and corner analysis thresholds. The error is calculated to be around 1%. This indicates that the design and fabrication were performed successfully.

Acknowledgments:

I/We acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Mateo Branion-Calles performed the measurements at The University of British Columbia. We

acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

References:

[1] R. J. Bojko, J. Li, L. He, T. Baehr-Jones, M. Hochberg, and Y. Aida, "Electron beam lithography writing strategies for low loss, high confinement silicon optical waveguides," J. Vacuum Sci. Technol. B 29, 06F309 (2011)

[2] Lukas Chrostowski, Michael Hochberg, chapter 12 in "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015

[3] http://siepic.ubc.ca/probestation, using Python code developed by Michael Caverley.

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